

Identity Verification – Dineo Malatjie

DRIVING LICENCE SADC SOUTH AFRICA

ZA

CARTA DE CONDUCAO
D MALATJIE

ID No.: 02/8708300844081 FEMALE

Birth: 20/05/1997 2A Restrictions: 1

Licence Number: 23550000925 No.: 1

Valid: 01/10/2023 - 10/10/2029

Issued: ZA

Code: B0

Vehicle restriction: 1

First issue: 01/10/2023

[Signature]



MSG

Meridian Compass

Your work. Our intelligence.
Always aligned.

An AI agent for employee policies,
travel and approvals

LINK TO REPO:

https://github.com/dineom-arch/Quantic_AI_Techniques_Project_Dineo_Malatjie

Agentic People & Travel Operations Assistant

Grounded policy guidance. Authorised context. Safe next actions.



Why this matters

A mobile, client-facing workforce means employee questions are part of delivering client value.

Employee questions become client-delivery questions.

A consulting workforce moves across countries, client assignments and approval structures. A simple question can depend on policy, operational context and the person authorised to decide.

1

Global assignments

Frequent travel and cross-border work

2

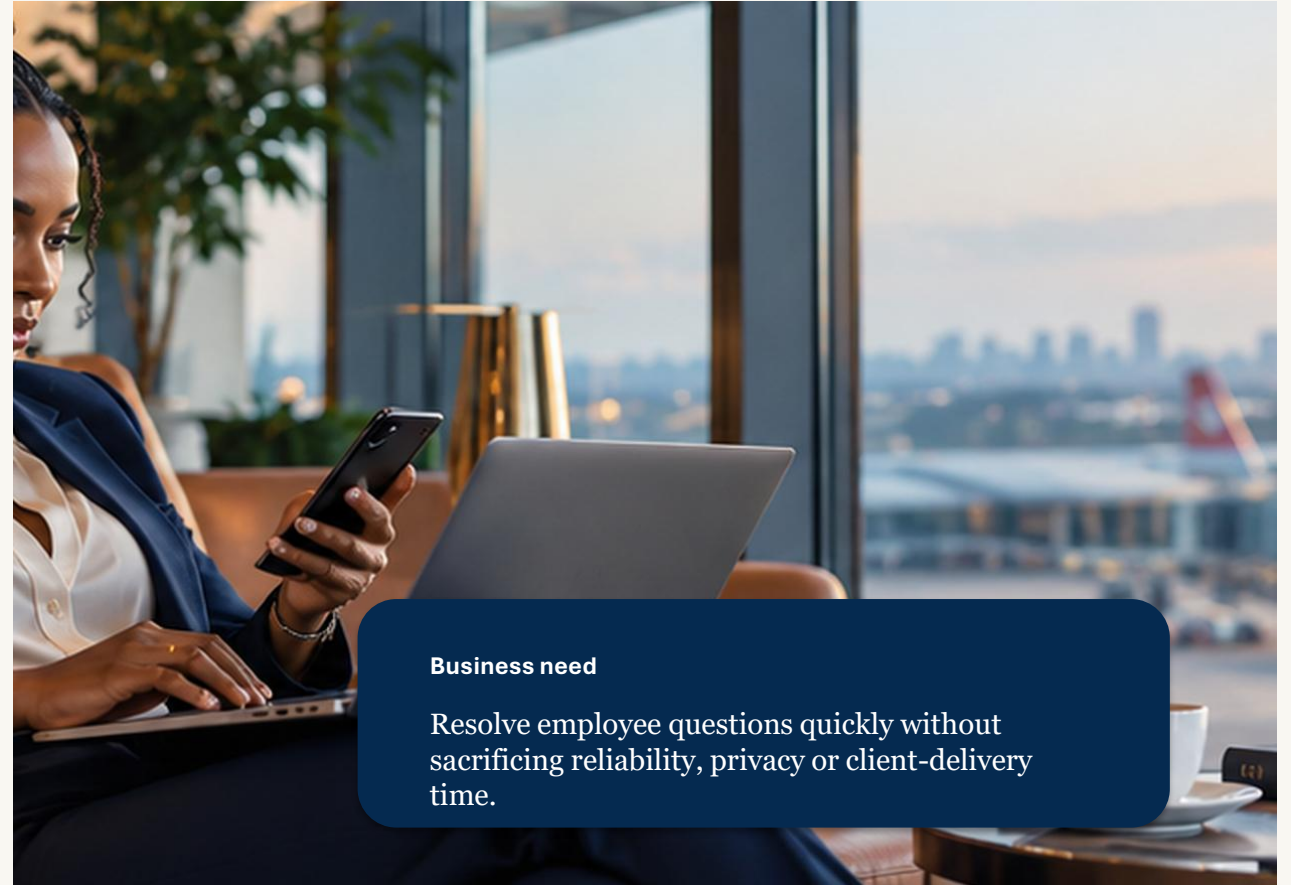
Client-facing teams

Fast answers without losing control

3

Complex policies

Multiple policies, roles and approvals



Business need

Resolve employee questions quickly without sacrificing reliability, privacy or client-delivery time.

The engineering problem was not just answering a question. It was proving why the answer was safe to use.

What the system had to do

Business requirements translated into explicit engineering controls.

BUSINESS REQUIREMENTS

Required capabilities

- 1

Identity
Know who the user is
- 2

Operational context
Know the current situation
- 3

Authoritative knowledge
Use approved MSG documents
- 4

Workflow tools
Read and update controlled systems
- 5

Privacy
Protect employee data
- 6

Grounding
Cite support for every answer
- 7

Safe boundaries
Know when to stop

Business requirements → Software requirements → Engineering controls → Verified behaviour

Design principle: if approved documents and authorised structured data do not support the answer, Meridian Compass does not know the answer.

How I engineered it

A controlled path from identity and intent to evidence, verification and response.

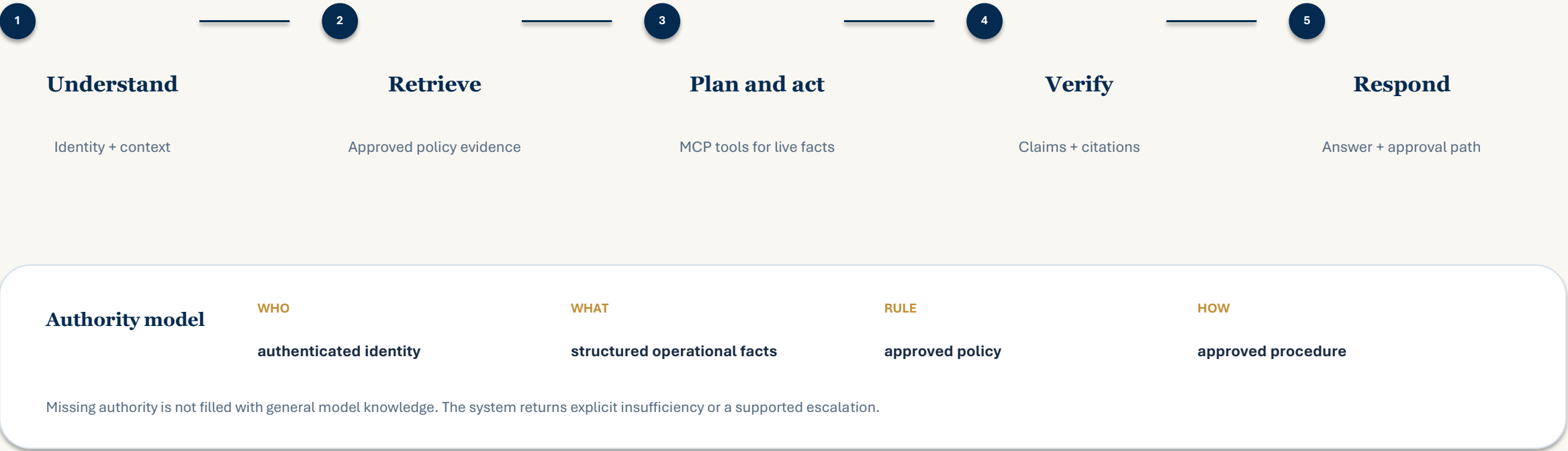


CONTROLLED EVIDENCE + ACTION BOUNDARY



How it works

From a question to a grounded, employee-specific answer.



RAG baseline k=5 | canonical Markdown corpus | FAISS IndexFlatIP | all-MiniLM-L6-v2 embeddings

Demo 1: Nairobi assignment PTO

A simple leave question becomes a multi-source approval decision.

USER QUESTION

“Can I take PTO on 15 and 16 October 2026?”

MERIDIAN COMPASS

Yes. You may submit the request.

You have 15 days available. The dates fall during your active Nairobi assignment. At runtime, the request gives 11 calendar days notice against the normal 14-day planned-PTO requirement.

Approval route: Amara Okafor, Engagement Manager.

Requestable does not mean approved.

What happened behind the answer

- 1

Identity + assignment

Naledi | Nairobi assignment active through 16 Oct
- 2

PTO balance

15 days available
- 3

Policy retrieval

Planned PTO requires at least 14 calendar days notice
- 4

Approval resolution

Active assignment route: Engagement Manager
- 5

Verification

Claims and citations checked before response

Production acceptance validated against live runtime context on 4 October 2026.

Demo 2: Personal travel extension

One request crosses travel policy, assignment context, cost ownership and approval routing.

SCENARIO

Extend the Nairobi return from 16 Oct to 20 Oct for personal travel.

BUSINESS RETURN **ZAR 8,400**

PROPOSED RETURN **ZAR 9,650**

EMPLOYEE INCREMENT **ZAR 1,250**

The assistant does not book real travel.

Only controlled mock actions after explicit confirmation.

1

Cost rule

MSG covers the business-equivalent authorised itinerary. Incremental personal cost is employee-funded.

2

Per diem

Per diem ends with the business period. Personal-extension days are not covered.

3

Approval

During an active assignment, the Engagement Manager approves before itinerary modification.

4

Working days

If the extension uses working days, a separate PTO decision is required.

Evaluation results

Controlled testing measured groundedness, agent behaviour, safety and latency.

90%

27 of 30 controlled tests passed

126 ms

p50 latency

356 ms

p95 latency

PASS

fail-safe ablation

PASS

live PTO production

Three failures were informative

EVAL006 Literal “eight” vs “8” mismatch

EVAL009 Prompt omitted required expense_id

EVAL010 Prompt omitted required expense_id

The system did not invent missing identifiers.

What this proved

Grounding and safe failure mattered more than forcing every test to pass.

The evaluation exposed contract quality, not just model quality.

Local deterministic latency: mean 168 ms | min 1 ms | max 865 ms

So what?

From fragmented employee effort to one controlled, contextual answer.

Current employee journey

- 1 Search policies
- 2 Check systems
- 3 Identify approver
- 4 Reconcile information
- 5 Contact enablement



With Meridian Compass

- 1 Ask one contextual question
- 2 Assemble authorised evidence
- 3 Resolve approval path
- 4 Verify claims and citations
- 5 Return next action



ENGINEERING VALUE

Grounded. Controlled. Observable.

Production learning: 512 MB OOM exposed in acceptance; service right-sized to 2 GB and the same build revalidated.

Meridian Compass turns a generic policy question into a grounded, employee-specific next action.

The LLM is one component of the controlled system, not the source of organisational truth.